LAPORAN PRAKTIKUM

APLIKASI BERBASIS PLATFORM

MODUL 6

Tugas 2 Praktikum

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PROGRAM STUDI S1 INFORMATIKA

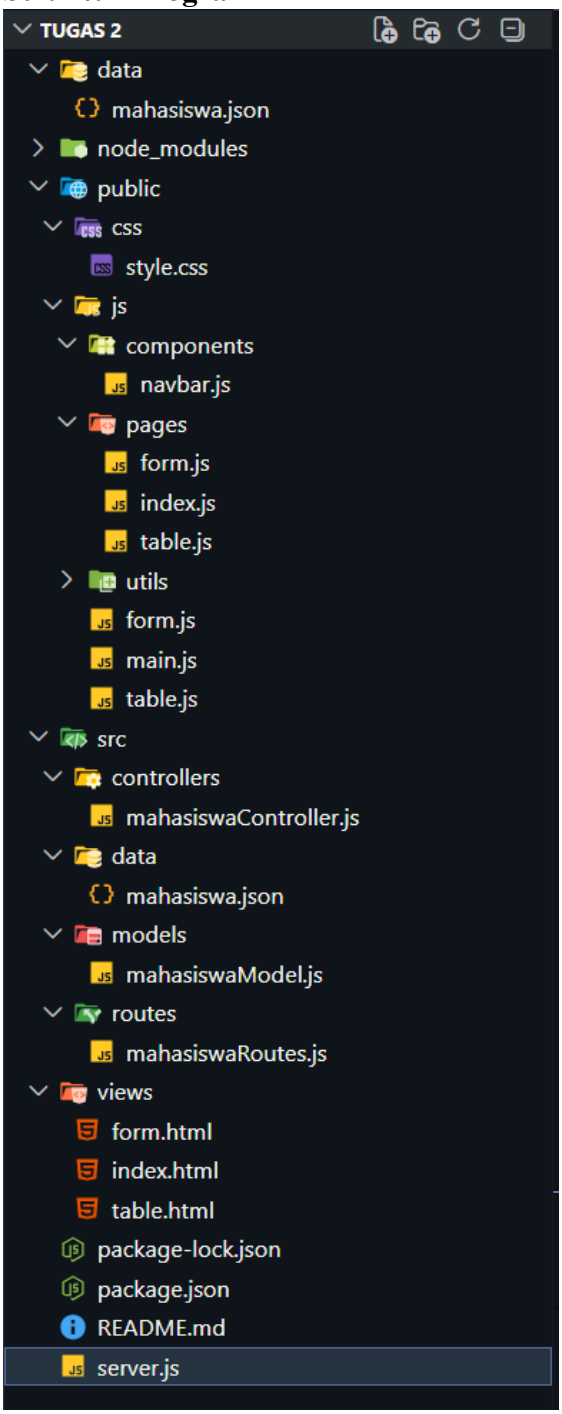
FAKULTAS INFORMATIKA

INSTITUT TEKNOLOGI TELKOM PURWOKERTO

2023/2024

Code Program

1. Struktur Program



Pada gambar diatas merupakan aplikasi web berbasis Node.js yang mengimplementasikan arsitektur *Model-View-Controller* (MVC) dengan pemisahan yang jelas antara *backend* (sisi peladen) dan *frontend* (sisi klien).

Berikut adalah rincian fungsionalitas dari setiap komponen dalam struktur tersebut:

1. Konfigurasi dan *Entry Point (Root Directory)*

   Pada tingkat utama (*root*), terdapat beberapa berkas esensial untuk menjalankan aplikasi:

   - server.js: Berfungsi sebagai berkas utama (*entry point*) untuk menginisialisasi dan menjalankan peladen (server) aplikasi.
   - package.json & package-lock.json: Berkas konfigurasi Node.js yang menyimpan informasi metadata proyek dan daftar *dependencies* (pustaka pihak ketiga) yang digunakan.
   - node_modules/: Direktori yang memuat seluruh pustaka dan package yang diinstal berdasarkan package.json.
   - README.md: Berkas markdown yang umumnya digunakan untuk menuliskan dokumentasi atau panduan penggunaan proyek.

2. Arsitektur *Backend* (Direktori src/)

   Direktori src/ memuat logika utama aplikasi sisi peladen (*backend*), yang secara terstruktur menerapkan pola MVC:

   - controllers/ (mahasiswaController.js): Mengatur alur logika antara Model dan View. Berfungsi memproses permintaan masuk (request) dan mengembalikan respons yang sesuai.
   - models/ (mahasiswaModel.js): Merepresentasikan struktur data dan menangani logika yang berhubungan langsung dengan manipulasi data mahasiswa.
   - routes/ (mahasiswaRoutes.js): Berfungsi untuk mendefinisikan titik akhir (endpoints/URL) dari aplikasi dan mengarahkannya ke fungsi controller yang tepat.
   - views/ (form.html, index.html, table.html): Menyimpan berkas kerangka antarmuka pengguna berbasis HTML.
   - data/ (mahasiswa.json): Bertindak sebagai database sementara (berbasis file JSON) untuk menyimpan dan membaca data mahasiswa dalam lingkup backend. (Catatan: Terdapat juga direktori data dengan fungsi serupa di tingkat root aplikasi).

3. Aset Statis *Frontend* (Direktori public/)

   Direktori public/ didedikasikan untuk menyimpan aset-aset statis yang akan disajikan ke peramban (browser) klien:

   - css/ (style.css): Menyimpan aturan *Cascading Style Sheets* untuk mengatur tampilan visual halaman web.
   - js/: Direktori skrip sisi klien (client-side) yang diorganisasi secara modular:
   - components/: Menyimpan skrip untuk elemen antarmuka yang dapat digunakan ulang (*reusable*), seperti navbar.js.
   - pages/: Menyimpan skrip spesifik untuk setiap halaman (form.js, index.js, table.js).
   - utils/: Umumnya berisi fungsi-fungsi bantuan (*helper*).

4. Source Code Program (Function View/)
Source code Indeks.html

```
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>SIAKAD — Sistem Informasi Akademik</title>
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.2/css/bo
otstrap.min.css">
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.0/css/all.min.css">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link
href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wg
ht@400;500;600;700;800&family=DM+Mono:wght@400;500&display=swap"
rel="stylesheet">
  <link rel="stylesheet" href="/css/style.css">
</head>
<body>

<!-- Hero Section -->
<div class="hero-section">
  <div class="hero-bg-shapes">
    <div class="shape shape-1"></div>
    <div class="shape shape-2"></div>
    <div class="shape shape-3"></div>
  </div>
  <div class="container py-5">
    <div class="row align-items-center">
      <div class="col-lg-7">
        <div class="hero-badge mb-3 animate-fade-up">
          <i class="fas fa-shield-halved me-2"></i>Platform
Akademik Terintegrasi
```

```
        </div>
        <h1 class="hero-title animate-fade-up delay-1">Sistem
Informasi <span class="text-accent">Akademik</span> Mahasiswa</h1>
        <p class="hero-desc animate-fade-up delay-2">Kelola data
akademik mahasiswa secara efisien dengan fitur CRUD lengkap,
pencarian cepat, dan laporan statistik real-time berbasis JSON.</p>
        <div class="d-flex gap-3 flex-wrap animate-fade-up delay-
3">
          <a href="/form" class="btn btn-hero-primary"><i
class="fas fa-user-plus me-2"></i>Daftarkan Mahasiswa</a>
          <a href="/data" class="btn btn-hero-outline"><i
class="fas fa-table me-2"></i>Lihat Rekap Data</a>
        </div>
      </div>
      <div class="col-lg-5 d-none d-lg-block text-center animate-
fade-up delay-2">
        <div class="hero-illustration">
          <i class="fas fa-user-graduate"></i>
        </div>
      </div>
    </div>
  </div>
</div>

<!-- Stats Cards -->
<div class="container my-5">
  <div class="row g-4" id="statsRow">
    <div class="col-sm-6 col-xl-3">
      <div class="stat-card stat-blue">
        <div class="stat-icon"><i class="fas fa-users"></i></div>
        <div class="stat-info">
          <div class="stat-label">Total Mahasiswa</div>
          <div class="stat-value" id="statTotal">—</div>
        </div>
        <div class="stat-bg-icon"><i class="fas fa-
users"></i></div>
      </div>
    </div>
    <div class="col-sm-6 col-xl-3">
      <div class="stat-card stat-green">
        <div class="stat-icon"><i class="fas fa-circle-
check"></i></div>
        <div class="stat-info">
          <div class="stat-label">Mahasiswa Aktif</div>
          <div class="stat-value" id="statAktif">—</div>
        </div>
        <div class="stat-bg-icon"><i class="fas fa-circle-
check"></i></div>
      </div>
    </div>
    <div class="col-sm-6 col-xl-3">
      <div class="stat-card stat-orange">
        <div class="stat-icon"><i class="fas fa-award"></i></div>
        <div class="stat-info">
          <div class="stat-label">Rata-rata IPK</div>
          <div class="stat-value" id="statIpk">—</div>
        </div>
        <div class="stat-bg-icon"><i class="fas fa-
award"></i></div>
      </div>
    </div>
```

```
    <div class="col-sm-6 col-xl-3">
      <div class="stat-card stat-purple">
        <div class="stat-icon"><i class="fas fa-sitemap"></i></div>
        <div class="stat-info">
          <div class="stat-label">Program Studi</div>
          <div class="stat-value" id="statProdi">—</div>
        </div>
        <div class="stat-bg-icon"><i class="fas fa-
sitemap"></i></div>
      </div>
    </div>
  </div>
</div>

<!-- Prodi Chart & Recent -->
<div class="container mb-5">
  <div class="row g-4">
    <div class="col-lg-5">
      <div class="card-modern h-100">
        <div class="card-modern-header">
          <h5><i class="fas fa-chart-bar me-2 text-
accent"></i>Distribusi Program Studi</h5>
        </div>
        <div class="card-modern-body" id="prodiDistChart">
          <div class="text-center py-4 text-muted"><i class="fas
fa-spinner fa-spin me-2"></i>Memuat data...</div>
        </div>
      </div>
    </div>
    <div class="col-lg-7">
      <div class="card-modern h-100">
        <div class="card-modern-header d-flex justify-content-
between align-items-center">
          <h5><i class="fas fa-clock-rotate-left me-2 text-
accent"></i>Pendaftaran Terbaru</h5>
          <a href="/data" class="btn-link-accent">Lihat semua <i
class="fas fa-arrow-right ms-1"></i></a>
        </div>
        <div class="card-modern-body" id="recentStudents">
          <div class="text-center py-4 text-muted"><i class="fas
fa-spinner fa-spin me-2"></i>Memuat data...</div>
        </div>
      </div>
    </div>
  </div>
</div>

<!-- Feature Highlight -->
<div class="container mb-5">
  <div class="row g-4">
    <div class="col-md-4">
      <div class="feature-card">
        <div class="feature-icon feature-blue"><i class="fas fa-
plus-circle"></i></div>
        <h6 class="feature-title">Tambah Data</h6>
        <p class="feature-desc">Daftarkan mahasiswa baru dengan
formulir validasi lengkap dan preview sebelum simpan.</p>
        <a href="/form" class="feature-link">Buka Form <i
class="fas fa-arrow-right ms-1"></i></a>
      </div>
    </div>
```

```
    <div class="col-md-4">
      <div class="feature-card">
        <div class="feature-icon feature-green"><i class="fas fa-
table-list"></i></div>
        <h6 class="feature-title">Kelola Data</h6>
        <p class="feature-desc">Tampilkan, edit, dan hapus data
mahasiswa dengan tabel interaktif berbasis JSON DataTable.</p>
        <a href="/data" class="feature-link">Lihat Tabel <i
class="fas fa-arrow-right ms-1"></i></a>
      </div>
    </div>
    <div class="col-md-4">
      <div class="feature-card">
        <div class="feature-icon feature-orange"><i class="fas fa-
chart-pie"></i></div>
        <h6 class="feature-title">Statistik Langsung</h6>
        <p class="feature-desc">Pantau distribusi program studi,
rata-rata IPK, dan jumlah mahasiswa aktif secara real-time.</p>
        <a href="#statsRow" class="feature-link">Lihat Statistik <i
class="fas fa-arrow-right ms-1"></i></a>
      </div>
    </div>
  </div>
</div>

<footer class="footer-section">
  <div class="container text-center">
    <p class="mb-0">© 2024 <strong>SIAKAD</strong> — Sistem
Informasi Akademik | Tugas 2 Praktikum Pemrograman Web</p>
  </div>
</footer>


<script
src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.7.1/jquery.min
.js"></script>
<script
src="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.2/js/boot
strap.bundle.min.js"></script>
<script
src="https://cdnjs.cloudflare.com/ajax/libs/toastr.js/2.1.4/toastr.
min.js"></script>
<script src="/js/components/navbar.js"></script>
<script src="/js/utils/api.js"></script>
<script src="/js/pages/index.js"></script>
</body>
</html>
```

Source code Form.html

```
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Daftarkan Mahasiswa — SIAKAD</title>
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.2/css/bo
otstrap.min.css">
```

```
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.0/css/all.min.css">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link
href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wg
ht@400;500;600;700;800&family=DM+Mono:wght@400;500&display=swap"
rel="stylesheet">
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/toastr.js/2.1.4/toastr
.min.css">
  <link rel="stylesheet" href="/css/style.css">
</head>
<body>

<div class="page-header">
  <div class="container-fluid px-4">
    <nav aria-label="breadcrumb">
      <ol class="breadcrumb mb-1">
        <li class="breadcrumb-item"><a href="/">Beranda</a></li>
        <li class="breadcrumb-item"><a href="/data">Rekap
Data</a></li>
        <li class="breadcrumb-item active">Daftarkan Mahasiswa</li>
      </ol>
    </nav>
    <h2 class="page-title mb-0"><i class="fas fa-user-plus me-
2"></i>Formulir Pendaftaran Mahasiswa</h2>
  </div>
</div>

<div class="container-fluid px-4 py-4">
  <div class="row justify-content-center">
    <div class="col-xl-8">
      <div id="formAlertArea"></div>

      <div class="card-modern">
        <div class="card-modern-header">
          <h5 class="mb-0"><i class="fas fa-file-pen me-2 text-
accent"></i>Data Pendaftaran Mahasiswa Baru</h5>
          <p class="text-muted small mb-0 mt-1">Isi semua field
bertanda <span class="text-danger fw-bold">*</span> yang wajib
diisi.</p>
        </div>
        <div class="card-modern-body p-4">
          <form id="studentForm" novalidate>

            <!-- Data Identitas -->
            <div class="form-section mb-4">
              <div class="form-section-title">
                <i class="fas fa-id-card me-2"></i>Identitas
Mahasiswa
              </div>
              <div class="row g-3">
                <div class="col-md-6">
                  <label class="form-label fw-600">NIM <span
class="text-danger">*</span></label>
                  <div class="input-group">
                    <span class="input-group-text"><i class="fas
fa-hashtag"></i></span>
                    <input type="text" class="form-control form-
control-modern" id="nim" name="nim"
```

```
                            placeholder="Contoh: 2022001"
maxlength="10">
                  </div>
                  <div class="invalid-feedback"
id="nimError"></div>
                  <div class="form-text">Nomor Induk Mahasiswa,
maks. 10 digit</div>
                </div>
                <div class="col-md-6">
                  <label class="form-label fw-600">Nama Lengkap
<span class="text-danger">*</span></label>
                  <div class="input-group">
                    <span class="input-group-text"><i class="fas
fa-user"></i></span>
                    <input type="text" class="form-control form-
control-modern" id="nama" name="nama"
                            placeholder="Nama lengkap sesuai KTP">
                  </div>
                  <div class="invalid-feedback"
id="namaError"></div>
                </div>
                <div class="col-md-12">
                  <label class="form-label fw-600">Alamat Email
<span class="text-danger">*</span></label>
                  <div class="input-group">
                    <span class="input-group-text"><i class="fas
fa-envelope"></i></span>
                    <input type="email" class="form-control form-
control-modern" id="email" name="email"
                            placeholder="nim@mahasiswa.ac.id">
                  </div>
                  <div class="invalid-feedback"
id="emailError"></div>
                </div>
              </div>
            </div>

            <!-- Data Akademik -->
            <div class="form-section mb-4">
              <div class="form-section-title">
                <i class="fas fa-book-open me-2"></i>Informasi
Akademik
              </div>
              <div class="row g-3">
                <div class="col-md-6">
                  <label class="form-label fw-600">Program Studi
<span class="text-danger">*</span></label>
                  <div class="input-group">
                    <span class="input-group-text"><i class="fas
fa-sitemap"></i></span>
                    <select class="form-select form-control-modern"
id="prodi" name="prodi">
                      <option value="">— Pilih Program Studi —
</option>
                      <option value="Teknik Informatika">Teknik
Informatika</option>
                      <option value="Sistem Informasi">Sistem
Informasi</option>
                      <option value="Teknik Komputer">Teknik
Komputer</option>
```

```
                        <option value="Ilmu Komputer">Ilmu
Komputer</option>
                        <option value="Rekayasa Perangkat
Lunak">Rekayasa Perangkat Lunak</option>
                      </select>
                    </div>
                    <div class="invalid-feedback"
id="prodiError"></div>
                  </div>
                  <div class="col-md-3">
                    <label class="form-label fw-600">Angkatan <span
class="text-danger">*</span></label>
                    <div class="input-group">
                      <span class="input-group-text"><i class="fas
fa-calendar"></i></span>
                      <select class="form-select form-control-modern"
id="angkatan" name="angkatan">
                        <option value="">Tahun</option>
                        <option>2020</option><option>2021</option><op
tion>2022</option>
                        <option>2023</option><option>2024</option>
                      </select>
                    </div>
                    <div class="invalid-feedback"
id="angkatanError"></div>
                  </div>
                  <div class="col-md-3">
                    <label class="form-label fw-600">Semester <span
class="text-danger">*</span></label>
                    <div class="input-group">
                      <span class="input-group-text"><i class="fas
fa-layer-group"></i></span>
                      <select class="form-select form-control-modern"
id="semester" name="semester">
                        <option value="">Pilih</option>
                        <option>1</option><option>2</option><option>3
</option><option>4</option>
                        <option>5</option><option>6</option><option>7
</option><option>8</option>
                        <option>9</option><option>10</option>
                      </select>
                    </div>
                    <div class="invalid-feedback"
id="semesterError"></div>
                  </div>
                  <div class="col-md-4">
                    <label class="form-label fw-600">IPK <span
class="text-danger">*</span></label>
                    <div class="input-group">
                      <span class="input-group-text"><i class="fas
fa-star-half-stroke"></i></span>
                      <input type="number" class="form-control form-
control-modern" id="ipk" name="ipk"
                             min="0" max="4" step="0.01"
placeholder="0.00">
                    </div>
                    <div class="invalid-feedback"
id="ipkError"></div>
                    <div class="form-text">Skala 0.00 – 4.00</div>
                  </div>
                  <div class="col-md-8">
```

```
                  <label class="form-label fw-600">Status
Akademik</label>
                  <div class="d-flex gap-3 mt-2 flex-wrap"
id="statusRadioGroup">
                    <div class="form-check form-check-custom">
                      <input class="form-check-input" type="radio"
name="status" id="statusAktif" value="Aktif" checked>
                      <label class="form-check-label"
for="statusAktif"><span class="dot dot-green"></span> Aktif</label>
                    </div>
                    <div class="form-check form-check-custom">
                      <input class="form-check-input" type="radio"
name="status" id="statusCuti" value="Cuti">
                      <label class="form-check-label"
for="statusCuti"><span class="dot dot-orange"></span> Cuti</label>
                    </div>
                    <div class="form-check form-check-custom">
                      <input class="form-check-input" type="radio"
name="status" id="statusLulus" value="Lulus">
                      <label class="form-check-label"
for="statusLulus"><span class="dot dot-blue"></span> Lulus</label>
                    </div>
                  </div>
                </div>
              </div>
            </div>

            <!-- Preview Card -->
            <div id="previewSection" class="preview-card mb-4"
style="display:none">
              <div class="preview-title"><i class="fas fa-eye me-
2"></i>Pratinjau Data Mahasiswa</div>
              <div class="row g-2 mt-1" id="previewContent"></div>
            </div>

            <!-- Action Buttons -->
            <div class="d-flex gap-3 justify-content-end flex-
wrap">
              <button type="button" class="btn btn-secondary"
id="btnReset">
                <i class="fas fa-rotate-left me-2"></i>Reset
              </button>
              <button type="button" class="btn btn-outline-primary-
custom" id="btnPreview">
                <i class="fas fa-eye me-2"></i>Pratinjau
              </button>
              <button type="submit" class="btn btn-primary-custom"
id="btnSubmit">
                <i class="fas fa-floppy-disk me-2"></i>Simpan
              </button>
            </div>
          </form>
        </div>
      </div>
    </div>

    <!-- Sidebar -->
    <div class="col-xl-4">
      <div class="card-modern sticky-top" style="top: 80px;">
        <div class="card-modern-header">
```

```
          <h6 class="mb-0"><i class="fas fa-circle-info me-2 text-
accent"></i>Petunjuk Pengisian</h6>
        </div>
        <div class="card-modern-body">
          <div class="guide-item">
            <div class="guide-icon"><i class="fas fa-
hashtag"></i></div>
            <div><strong>NIM</strong> — Nomor Induk Mahasiswa unik,
hanya angka, maks. 10 digit.</div>
          </div>
          <div class="guide-item">
            <div class="guide-icon"><i class="fas fa-
sitemap"></i></div>
            <div><strong>Program Studi</strong> — Pilih program
studi yang sedang ditempuh.</div>
          </div>
          <div class="guide-item">
            <div class="guide-icon"><i class="fas fa-
calendar"></i></div>
            <div><strong>Angkatan</strong> — Tahun pertama kali
mahasiswa terdaftar.</div>
          </div>
          <div class="guide-item">
            <div class="guide-icon"><i class="fas fa-star-half-
stroke"></i></div>
            <div><strong>IPK</strong> — Indeks Prestasi Kumulatif
dalam skala 0.00–4.00.</div>
          </div>
          <div class="guide-item">
            <div class="guide-icon"><i class="fas fa-circle-
dot"></i></div>
            <div><strong>Status</strong> — Pilih status akademik
mahasiswa saat ini.</div>
          </div>
          <hr class="my-3">
          <div class="d-grid">
            <a href="/data" class="btn btn-outline-secondary btn-
sm">
              <i class="fas fa-list me-2"></i>Lihat Semua Data
            </a>
          </div>
        </div>
      </div>
    </div>
  </div>
</div>

<footer class="footer-section">
  <div class="container text-center">
    <p class="mb-0">© 2024 <strong>SIAKAD</strong> — Sistem
Informasi Akademik | Tugas 2 Praktikum Pemrograman Web</p>
  </div>
</footer>

<script
src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.7.1/jquery.min
.js"></script>
<script
src="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.2/js/boot
strap.bundle.min.js"></script>
```

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/jquery-
validate/1.19.5/jquery.validate.min.js"></script>
<script
src="https://cdnjs.cloudflare.com/ajax/libs/toastr.js/2.1.4/toastr.
min.js"></script>
<script src="/js/components/navbar.js"></script>
<script src="/js/utils/api.js"></script>
<script src="/js/pages/form.js"></script>
</body>
</html>
```

Source code table.html

```
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Rekap Data Mahasiswa — SIAKAD</title>
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.2/css/bo
otstrap.min.css">
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.0/css/all.min.css">
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/datatables/1.10.21/css
/dataTables.bootstrap5.min.css">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link
href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wg
ht@400;500;600;700;800&family=DM+Mono:wght@400;500&display=swap"
rel="stylesheet">
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/toastr.js/2.1.4/toastr
.min.css">
  <link rel="stylesheet" href="/css/style.css">
</head>
<body>

<div class="page-header">
  <div class="container-fluid px-4">
    <div class="d-flex justify-content-between align-items-center
flex-wrap gap-3">
      <div>
        <nav aria-label="breadcrumb">
          <ol class="breadcrumb mb-1">
            <li class="breadcrumb-item"><a
href="/">Beranda</a></li>
            <li class="breadcrumb-item active">Rekap Data
Mahasiswa</li>
          </ol>
        </nav>
        <h2 class="page-title mb-0"><i class="fas fa-table me-
2"></i>Rekap Data Mahasiswa</h2>
      </div>
      <a href="/form" class="btn btn-primary-custom">
        <i class="fas fa-user-plus me-2"></i>Daftarkan Mahasiswa
      </a>
    </div>
```

```
  </div>
</div>

<div class="container-fluid px-4 py-4">
  <div id="alertArea"></div>

  <div class="card-modern">
    <div class="card-modern-header d-flex justify-content-between
align-items-center flex-wrap gap-2">
      <h5 class="mb-0"><i class="fas fa-database me-2 text-
accent"></i>Tabel Data Mahasiswa</h5>
      <div class="d-flex gap-2">
        <button class="btn btn-sm btn-outline-secondary"
id="btnRefresh" title="Muat ulang data">
          <i class="fas fa-arrows-rotate me-1"></i>Muat Ulang
        </button>
      </div>
    </div>
    <div class="card-modern-body p-0">
      <div class="table-responsive">
        <table id="mahasiswaTable" class="table table-hover align-
middle mb-0 w-100">
          <thead>
            <tr>
              <th width="50">No</th>
              <th>NIM</th>
              <th>Nama Mahasiswa</th>
              <th>Program Studi</th>
              <th>Angkatan</th>
              <th>Sem</th>
              <th>IPK</th>
              <th>Status</th>
              <th width="110">Aksi</th>
            </tr>
          </thead>
          <tbody></tbody>
        </table>
      </div>
    </div>
  </div>
</div>

<!-- Modal Edit -->
<div class="modal fade" id="editModal" tabindex="-1">
  <div class="modal-dialog modal-lg modal-dialog-centered">
    <div class="modal-content modal-modern">
      <div class="modal-header">
        <h5 class="modal-title"><i class="fas fa-pen-to-square me-
2"></i>Edit Data Mahasiswa</h5>
        <button type="button" class="btn-close btn-close-white"
data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body p-4">
        <div id="editAlertArea"></div>
        <div class="row g-3">
          <input type="hidden" id="editId">
          <div class="col-md-6">
            <label class="form-label fw-600">NIM <span class="text-
danger">*</span></label>
            <input type="text" class="form-control form-control-
modern" id="editNim" placeholder="Contoh: 2022001">
```

```
          </div>
          <div class="col-md-6">
            <label class="form-label fw-600">Nama Lengkap <span
class="text-danger">*</span></label>
            <input type="text" class="form-control form-control-
modern" id="editNama" placeholder="Nama lengkap mahasiswa">
          </div>
          <div class="col-md-6">
            <label class="form-label fw-600">Program Studi <span
class="text-danger">*</span></label>
            <select class="form-select form-control-modern"
id="editProdi">
              <option value="">Pilih Program Studi</option>
              <option>Teknik Informatika</option>
              <option>Sistem Informasi</option>
              <option>Teknik Komputer</option>
              <option>Ilmu Komputer</option>
              <option>Rekayasa Perangkat Lunak</option>
            </select>
          </div>
          <div class="col-md-3">
            <label class="form-label fw-600">Angkatan <span
class="text-danger">*</span></label>
            <select class="form-select form-control-modern"
id="editAngkatan">
              <option value="">Tahun</option>
              <option>2020</option><option>2021</option><option>202
2</option>
              <option>2023</option><option>2024</option>
            </select>
          </div>
          <div class="col-md-3">
            <label class="form-label fw-600">Semester <span
class="text-danger">*</span></label>
            <select class="form-select form-control-modern"
id="editSemester">
              <option value="">Pilih</option>
              <option>1</option><option>2</option><option>3</option
><option>4</option>
              <option>5</option><option>6</option><option>7</option
><option>8</option>
              <option>9</option><option>10</option>
            </select>
          </div>
          <div class="col-md-4">
            <label class="form-label fw-600">IPK <span class="text-
danger">*</span></label>
            <input type="number" class="form-control form-control-
modern" id="editIpk" min="0" max="4" step="0.01"
placeholder="0.00">
          </div>
          <div class="col-md-8">
            <label class="form-label fw-600">Alamat Email <span
class="text-danger">*</span></label>
            <input type="email" class="form-control form-control-
modern" id="editEmail" placeholder="nim@mahasiswa.ac.id">
          </div>
          <div class="col-md-4">
            <label class="form-label fw-600">Status
Akademik</label>
```

```
            <select class="form-select form-control-modern"
id="editStatus">
              <option value="Aktif">Aktif</option>
              <option value="Cuti">Cuti</option>
              <option value="Lulus">Lulus</option>
              <option value="Keluar">Keluar</option>
            </select>
          </div>
        </div>
      </div>
      <div class="modal-footer">
        <button type="button" class="btn btn-secondary" data-bs-
dismiss="modal">Batal</button>
        <button type="button" class="btn btn-primary-custom"
id="btnSaveEdit">
          <i class="fas fa-floppy-disk me-2"></i>Simpan Perubahan
        </button>
      </div>
    </div>
  </div>
</div>

<!-- Modal Konfirmasi Hapus -->
<div class="modal fade" id="deleteModal" tabindex="-1">
  <div class="modal-dialog modal-dialog-centered modal-sm">
    <div class="modal-content modal-modern">
      <div class="modal-header border-0">
        <h5 class="modal-title text-danger"><i class="fas fa-
triangle-exclamation me-2"></i>Konfirmasi Hapus</h5>
        <button type="button" class="btn-close btn-close-white"
data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body text-center py-3">
        <div class="delete-icon-wrap mb-3"><i class="fas fa-trash-
can"></i></div>
        <p class="mb-1">Yakin hapus data mahasiswa:</p>
        <strong id="deleteStudentName" class="text-
accent"></strong>
        <p class="text-muted small mt-2 mb-0">Data yang dihapus
tidak bisa dikembalikan.</p>
      </div>
      <div class="modal-footer border-0 justify-content-center gap-
2">
        <button type="button" class="btn btn-secondary" data-bs-
dismiss="modal">Batal</button>
        <button type="button" class="btn btn-danger"
id="btnConfirmDelete">
          <i class="fas fa-trash-can me-1"></i>Hapus
        </button>
      </div>
    </div>
  </div>
</div>

<footer class="footer-section">
  <div class="container text-center">
    <p class="mb-0">© 2024 <strong>SIAKAD</strong> — Sistem
Informasi Akademik | Tugas 2 Praktikum Pemrograman Web</p>
  </div>
</footer>
```

```
<script
src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.7.1/jquery.min
.js"></script>
<script
src="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.2/js/boot
strap.bundle.min.js"></script>
<script
src="https://cdnjs.cloudflare.com/ajax/libs/datatables/1.10.21/js/j
query.dataTables.min.js"></script>
<script
src="https://cdnjs.cloudflare.com/ajax/libs/datatables/1.10.21/js/d
ataTables.bootstrap5.min.js"></script>
<script
src="https://cdnjs.cloudflare.com/ajax/libs/toastr.js/2.1.4/toastr.
min.js"></script>
<script src="/js/components/navbar.js"></script>
<script src="/js/utils/api.js"></script>
<script src="/js/pages/table.js"></script>
</body>
</html>
```

5. **Source Code Program (Function src/)**
   Source code mahasiswaController.js(Controller)

```
const { v4: uuidv4 } = require('uuid');
const model = require('../models/mahasiswaModel');

// GET /api/mahasiswa
function getAll(req, res) {
  res.json({ data: model.getAll() });
}

// GET /api/mahasiswa/:id
function getById(req, res) {
  const mhs = model.getById(req.params.id);
  if (!mhs) return res.status(404).json({ error: 'Data tidak
ditemukan' });
  res.json(mhs);
}

// POST /api/mahasiswa
function create(req, res) {
  const { nim, nama, prodi, angkatan, semester, ipk, email, status
} = req.body;

  if (!nim || !nama || !prodi || !angkatan || !semester || !ipk ||
!email)
    return res.status(400).json({ error: 'Semua field wajib diisi'
});

  if (model.nimExists(nim))
    return res.status(400).json({ error: 'NIM sudah terdaftar' });

  const newMhs = {
    id: uuidv4(), nim, nama, prodi,
    angkatan: parseInt(angkatan),
    semester:  parseInt(semester),
    ipk:       parseFloat(ipk),
    email,
    status:    status || 'Aktif',
```

```
    createdAt: new Date().toISOString()
  };

  model.create(newMhs);
  res.status(201).json({ success: true, message: 'Data berhasil
ditambahkan', data: newMhs });
}

// PUT /api/mahasiswa/:id
function update(req, res) {
  const { nim, nama, prodi, angkatan, semester, ipk, email, status
} = req.body;

  if (model.nimExists(nim, req.params.id))
    return res.status(400).json({ error: 'NIM sudah digunakan
mahasiswa lain' });

  const updated = model.update(req.params.id, {
    nim, nama, prodi,
    angkatan:  parseInt(angkatan),
    semester:  parseInt(semester),
    ipk:       parseFloat(ipk),
    email,
    status:    status || 'Aktif',
    updatedAt: new Date().toISOString()
  });

  if (!updated) return res.status(404).json({ error: 'Data tidak
ditemukan' });
  res.json({ success: true, message: 'Data berhasil diperbarui',
data: updated });
}

// DELETE /api/mahasiswa/:id
function remove(req, res) {
  const deleted = model.remove(req.params.id);
  if (!deleted) return res.status(404).json({ error: 'Data tidak
ditemukan' });
  res.json({ success: true, message: `Data ${deleted.nama} berhasil
dihapus` });
}

// GET /api/stats
function getStats(req, res) {
  const list  = model.getAll();
  const total = list.length;
  const aktif = list.filter(s => s.status === 'Aktif').length;
  const avgIpk = total > 0
    ? (list.reduce((sum, s) => sum + s.ipk, 0) / total).toFixed(2)
    : 0;

  const prodiCount = {};
  list.forEach(s => { prodiCount[s.prodi] = (prodiCount[s.prodi] ||
0) + 1; });

  res.json({ total, aktif, avgIpk, prodiCount });
}

module.exports = { getAll, getById, create, update, remove,
getStats };
```

Source code Mahasiswa.json(Data Dummy biar tidak kosong)

```
{
  "mahasiswa": [
    {
      "id": "a1b2c3d4-0001-0001-0001-000000000001",
      "nim": "2022001",
      "nama": "Budi Santoso",
      "prodi": "Teknik Informatika",
      "angkatan": 2022,
      "semester": 5,
      "ipk": 3.75,
      "email": "budi@student.ac.id",
      "status": "Aktif",
      "createdAt": "2024-01-10T08:00:00.000Z"
    },
    {
      "id": "a1b2c3d4-0002-0002-0002-000000000002",
      "nim": "2021002",
      "nama": "Siti Rahayu",
      "prodi": "Sistem Informasi",
      "angkatan": 2021,
      "semester": 7,
      "ipk": 3.92,
      "email": "siti@student.ac.id",
      "status": "Aktif",
      "createdAt": "2024-01-11T09:00:00.000Z"
    },
    {
      "id": "a1b2c3d4-0003-0003-0003-000000000003",
      "nim": "2020003",
      "nama": "Ahmad Fauzi",
      "prodi": "Teknik Komputer",
      "angkatan": 2020,
      "semester": 9,
      "ipk": 3.45,
      "email": "ahmad@student.ac.id",
      "status": "Lulus",
      "createdAt": "2024-01-12T10:00:00.000Z"
    },
    {
      "id": "21ad2032-e71d-4358-8b43-4aaca8a2ddaa",
      "nim": "2311102262",
      "nama": "Aji Noto Sutrisno",
      "prodi": "Teknik Informatika",
      "angkatan": 2023,
      "semester": 6,
      "ipk": 3.83,
      "email": "ajinotosutrisno212@gmail.com",
      "status": "Aktif",
      "createdAt": "2026-04-05T19:15:09.910Z"
    }
  ]
}
```

Source code mahasiswaModel.js

```
const fs   = require('fs');
```

```
const path = require('path');

const DATA_FILE = path.join(__dirname, '../data/mahasiswa.json');

function readData() {
  if (!fs.existsSync(DATA_FILE)) {
    const initial = { mahasiswa: [] };
    fs.writeFileSync(DATA_FILE, JSON.stringify(initial, null, 2));
    return initial;
  }
  return JSON.parse(fs.readFileSync(DATA_FILE, 'utf8'));
}

function writeData(data) {
  fs.writeFileSync(DATA_FILE, JSON.stringify(data, null, 2));
}

function getAll() {
  return readData().mahasiswa;
}

function getById(id) {
  return readData().mahasiswa.find(s => s.id === id) || null;
}

function create(mhs) {
  const db = readData();
  db.mahasiswa.push(mhs);
  writeData(db);
  return mhs;
}

function update(id, updated) {
  const db = readData();
  const index = db.mahasiswa.findIndex(s => s.id === id);
  if (index === -1) return null;
  db.mahasiswa[index] = { ...db.mahasiswa[index], ...updated };
  writeData(db);
  return db.mahasiswa[index];
}

function remove(id) {
  const db = readData();
  const index = db.mahasiswa.findIndex(s => s.id === id);
  if (index === -1) return null;
  const deleted = db.mahasiswa.splice(index, 1)[0];
  writeData(db);
  return deleted;
}

function nimExists(nim, excludeId = null) {
  return readData().mahasiswa.some(s => s.nim === nim && s.id !==
excludeId);
}

module.exports = { getAll, getById, create, update, remove,
nimExists };
```

Source code mahasiswaRoute.js (routes)

```
const express    = require('express');
```

```
const router     = express.Router();
const controller = require('../controllers/mahasiswaController');

router.get('/',        controller.getAll);
router.get('/stats',   controller.getStats);
router.get('/:id',     controller.getById);
router.post('/',       controller.create);
router.put('/:id',     controller.update);
router.delete('/:id',  controller.remove);

module.exports = router;
```

6. **Source Code Program (Function js/)**
   Source code navbar.js(Component)

```
// components/navbar.js — Komponen navbar yang dapat digunakan
ulang

const Navbar = {
  // Render HTML navbar ke dalam selector yang diberikan
  render(selector = 'body') {
    const html = `
      <nav class="navbar navbar-expand-lg navbar-dark sticky-top"
id="mainNav">
        <div class="container-fluid px-4">
          <a class="navbar-brand d-flex align-items-center gap-2"
href="/">
            <div class="brand-icon"><i class="fas fa-
university"></i></div>
            <span class="brand-text">SIAKAD</span>
          </a>
          <button class="navbar-toggler border-0" type="button"
                  data-bs-toggle="collapse" data-bs-
target="#navMenu">
            <span class="navbar-toggler-icon"></span>
          </button>
          <div class="collapse navbar-collapse" id="navMenu">
            <ul class="navbar-nav ms-auto gap-1">
              <li class="nav-item">
                <a class="nav-link" href="/"><i class="fas fa-house
me-1"></i>Beranda</a>
              </li>
              <li class="nav-item">
                <a class="nav-link" href="/data"><i class="fas fa-
table me-1"></i>Rekap Data</a>
              </li>
              <li class="nav-item">
                <a class="nav-link" href="/form"><i class="fas fa-
user-plus me-1"></i>Daftarkan</a>
              </li>
            </ul>
          </div>
        </div>
      </nav>`;

    $(selector).prepend(html);
    this.setActiveLink();
  },

  // Tandai link yang aktif berdasarkan URL saat ini
  setActiveLink() {
    const currentPath = window.location.pathname;
```

```
    $('.navbar-nav .nav-link').each(function () {
      const href = $(this).attr('href');
      const isActive = href === currentPath ||
                       (href === '/' && currentPath === '/');
      $(this).toggleClass('active', isActive);
    });
  }
};
```

Source code Form.js(Pages js)

```
// pages/form.js — Validasi form & submit tambah mahasiswa

$(document).ready(function () {
  // Render navbar
  Navbar.render('body');

  // ── jQuery Validate
──────────────────────────────────────────
  $('#studentForm').validate({
    rules: {
      nim:      { required: true, minlength: 5, maxlength: 10 },
      nama:     { required: true, minlength: 3 },
      email:    { required: true, email: true },
      prodi:    { required: true },
      angkatan: { required: true },
      semester: { required: true },
      ipk:      { required: true, min: 0, max: 4, number: true }
    },
    messages: {
      nim:      { required: 'NIM wajib diisi', minlength: 'Minimal
5 karakter', maxlength: 'Maksimal 10 karakter' },
      nama:     { required: 'Nama wajib diisi', minlength: 'Minimal
3 karakter' },
      email:    { required: 'Email wajib diisi', email: 'Format
email tidak valid' },
      prodi:    { required: 'Program studi wajib dipilih' },
      angkatan: { required: 'Angkatan wajib dipilih' },
      semester: { required: 'Semester wajib dipilih' },
      ipk:      { required: 'IPK wajib diisi', min: 'Minimal 0.00',
max: 'Maksimal 4.00', number: 'Harus berupa angka' }
    },
    errorElement: 'div',
    errorClass:   'invalid-feedback',
    highlight:    el => $(el).addClass('is-
invalid').removeClass('is-valid'),
    unhighlight:  el => $(el).removeClass('is-
invalid').addClass('is-valid'),
    errorPlacement(error, element) {
      element.parent().hasClass('input-group')
        ? error.insertAfter(element.parent())
        : error.insertAfter(element);
    },
    submitHandler: () => submitForm()
  });

  // ── Submit Form (CREATE)
──────────────────────────────────────────
  function submitForm() {
    const $btn = $('#btnSubmit');
```

```
    $btn.html('<i class="fas fa-spinner fa-spin me-
2"></i>Menyimpan...').prop('disabled', true);
    $('#formAlertArea').html('');

    const payload = {
      nim:      $('#nim').val().trim(),
      nama:     $('#nama').val().trim(),
      email:    $('#email').val().trim(),
      prodi:    $('#prodi').val(),
      angkatan: $('#angkatan').val(),
      semester: $('#semester').val(),
      ipk:      $('#ipk').val(),
      status:   $('input[name="status"]:checked').val() || 'Aktif'
    };

    api.create(payload)
      .done(function (res) {
        toastr.success(res.message);
        $('#formAlertArea').html(`
          <div class="alert alert-success d-flex align-items-center
gap-2 rounded-3">
            <i class="fas fa-circle-check"></i>
            <div><strong>Berhasil!</strong> Mahasiswa
<strong>${res.data.nama}</strong> telah terdaftar.
            <a href="/data" class="alert-link">Lihat rekap data
→</a></div>
          </div>`);
        resetForm();
        $('#previewSection').slideUp(200);
      })
      .fail(function (xhr) {
        const msg = xhr.responseJSON?.error || 'Terjadi kesalahan
pada server';
        toastr.error(msg);
        $('#formAlertArea').html(`
          <div class="alert alert-danger d-flex align-items-center
gap-2 rounded-3">
            <i class="fas fa-circle-xmark"></i>
            <div><strong>Gagal!</strong> ${msg}</div>
          </div>`);
      })
      .always(() => $btn.html('<i class="fas fa-floppy-disk me-
2"></i>Simpan').prop('disabled', false));
  }

  // ── Tombol Pratinjau
──────────────────────────────────────────────
  $('#btnPreview').on('click', function () {
    if (!$('#studentForm').valid()) {
      toastr.warning('Lengkapi semua field yang wajib diisi
terlebih dahulu');
      return;
    }
    const ipk = parseFloat($('#ipk').val()).toFixed(2);
    $('#previewContent').html(`
      <div class="col-6 col-md-3">
        <div class="preview-item-label">NIM</div>
        <div class="preview-item-val">${$('#nim').val()}</div>
      </div>
      <div class="col-6 col-md-5">
        <div class="preview-item-label">Nama</div>
```

```
        <div class="preview-item-val">${$('#nama').val()}</div>
      </div>
      <div class="col-12 col-md-4">
        <div class="preview-item-label">Email</div>
        <div class="preview-item-val" style="font-
size:.8rem">${$('#email').val()}</div>
      </div>
      <div class="col-6 col-md-4">
        <div class="preview-item-label">Program Studi</div>
        <div class="preview-item-val">${$('#prodi').val()}</div>
      </div>
      <div class="col-6 col-md-2">
        <div class="preview-item-label">Angkatan</div>
        <div class="preview-item-val">${$('#angkatan').val()}</div>
      </div>
      <div class="col-6 col-md-2">
        <div class="preview-item-label">Semester</div>
        <div class="preview-item-val">Sem-
${$('#semester').val()}</div>
      </div>
      <div class="col-6 col-md-2">
        <div class="preview-item-label">IPK</div>
        <div class="preview-item-val" style="color:var(--
warning)">${ipk}</div>
      </div>
      <div class="col-6 col-md-2">
        <div class="preview-item-label">Status</div>
        <div class="preview-item-
val">${$('input[name="status"]:checked').val() || 'Aktif'}</div>
      </div>`);
    $('#previewSection').slideDown(300);
    $('html, body').animate({ scrollTop:
$('#previewSection').offset().top - 100 }, 500);
  });

  // ── Tombol Reset
──────────────────────────────────────────────
  $('#btnReset').on('click', function () {
    resetForm();
    $('#previewSection').slideUp(200);
    $('#formAlertArea').html('');
    toastr.info('Form berhasil direset');
  });

  function resetForm() {
    $('#studentForm')[0].reset();
    $('#studentForm .is-invalid, #studentForm .is-
valid').removeClass('is-invalid is-valid');
    $('#statusAktif').prop('checked', true);
  }

  // ── Input Filtering
─────────────────────────────────────────────
  $('#nim').on('input', function () {
    $(this).val($(this).val().replace(/\D/g, ''));
  });

  $('#ipk').on('input', function () {
    const v = parseFloat($(this).val());
    if (v > 4) $(this).val(4);
    if (v < 0) $(this).val(0);
```

```
  });

  $('input, select').on('focus', function () {
    $(this).closest('.input-group').find('.input-group-
text').addClass('ig-focused');
  }).on('blur', function () {
    $(this).closest('.input-group').find('.input-group-
text').removeClass('ig-focused');
  });
});
```

Source code index.js(pages js)

```
// pages/index.js — Script halaman Beranda

$(document).ready(function () {
  // Render navbar (dari component)
  Navbar.render('body');

  // Konfigurasi toastr
  toastr.options = {
    closeButton: true,
    progressBar: true,
    positionClass: 'toast-top-right',
    timeOut: 3500
  };

  // ── Muat Statistik
──────────────────────────────────────────────────
  api.getStats().done(function (data) {
    $('#statTotal').text(data.total);
    $('#statAktif').text(data.aktif);
    $('#statIpk').text(data.avgIpk);
    $('#statProdi').text(Object.keys(data.prodiCount).length);

    // Grafik distribusi prodi
    const colors = ['#1565c0', '#7209b7', '#e65100', '#2e7d32',
'#006064'];
    const total  = data.total || 1;
    let html = '';

    Object.entries(data.prodiCount).forEach(function ([prodi,
count], i) {
      const pct = Math.round((count / total) * 100);
      html += `
        <div class="mb-3">
          <div class="d-flex justify-content-between mb-1">
            <span class="fw-500 small">${prodi}</span>
            <span class="badge-count">${count} mhs</span>
          </div>
          <div class="progress-bar-wrap">
            <div class="progress-bar-fill"
style="width:${pct}%;background:${colors[i %
colors.length]}"></div>
          </div>
        </div>`;
    });

    $('#prodiDistChart').html(
      html || '<p class="text-muted text-center py-3">Belum ada
data</p>'
```

```
    );
  });

  // ── Muat Pendaftaran Terbaru
──────────────────────────────────────
  api.getAll().done(function (res) {
    const recent = res.data.slice(-4).reverse();

    if (recent.length === 0) {
      $('#recentStudents').html(
        '<p class="text-muted text-center py-3">Belum ada mahasiswa
terdaftar</p>'
      );
      return;
    }

    const getStatusClass = s =>
      ({ Aktif: 'badge-aktif', Cuti: 'badge-cuti', Lulus: 'badge-
lulus', Keluar: 'badge-keluar' })[s] || 'badge-aktif';

    let html = '<div class="student-list">';
    recent.forEach(function (s) {
      const initials = s.nama.split(' ').map(n =>
n[0]).join('').substring(0, 2).toUpperCase();
      html += `
        <div class="student-list-item">
          <div class="student-avatar">${initials}</div>
          <div class="student-info">
            <div class="student-name">${s.nama}</div>
            <div class="student-meta">${s.nim} · ${s.prodi} ·
Angkatan ${s.angkatan}</div>
          </div>
          <div class="text-end">
            <span class="ipk-badge">IPK ${s.ipk.toFixed(2)}</span>
            <div class="mt-1"><span class="status-badge
${getStatusClass(s.status)}">${s.status}</span></div>
          </div>
        </div>`;
    });
    html += '</div>';
    $('#recentStudents').html(html);
  });
});
```

Source code table.js(pages js)

```
// pages/table.js — DataTable CRUD (Read, Update, Delete)

let mahasiswaTable;
let deleteTargetId = null;

$(document).ready(function () {
  // Render navbar
  Navbar.render('body');

  // ── Inisialisasi DataTable
──────────────────────────────────────
  mahasiswaTable = $('#mahasiswaTable').DataTable({
    ajax:    { url: '/api/mahasiswa', type: 'GET', dataSrc: 'data'
},
    columns: [
```

```
      {
        data: null,
        orderable: false,
        render: (data, type, row, meta) => `<span class="nim-
mono">${meta.row + 1}</span>`
      },
      { data: 'nim',      render: d => `<span class="nim-
mono">${d}</span>` },
      { data: 'nama',     render: d => `<strong>${d}</strong>` },
      { data: 'prodi' },
      { data: 'angkatan', render: d => `<span class="nim-
mono">${d}</span>` },
      { data: 'semester', render: d => `<span class="nim-mono">Sem-
${d}</span>` },
      {
        data: 'ipk',
        render(d) {
          const v   = parseFloat(d);
          const cls = v >= 3.5 ? 'ipk-high' : v >= 3.0 ? 'ipk-mid'
: 'ipk-low';
          return `<span class="ipk-display
${cls}">${v.toFixed(2)}</span>`;
        }
      },
      {
        data: 'status',
        render(d) {
          const map = { Aktif:'badge-aktif', Cuti:'badge-cuti',
Lulus:'badge-lulus', Keluar:'badge-keluar' };
          return `<span class="status-badge ${map[d] || 'badge-
aktif'}">${d}</span>`;
        }
      },
      {
        data: null,
        orderable: false,
        render: (data, type, row) => `
          <div class="d-flex gap-1">
            <button class="btn-action btn-edit"   data-
id="${row.id}" title="Edit data">
              <i class="fas fa-pen"></i>
            </button>
            <button class="btn-action btn-delete" data-
id="${row.id}" data-nama="${row.nama}" title="Hapus data">
              <i class="fas fa-trash-can"></i>
            </button>
          </div>`
      }
    ],
    language: {
      search:      'Cari:',
      lengthMenu:  'Tampilkan _MENU_ data',
      info:        'Menampilkan _START_-_END_ dari _TOTAL_
mahasiswa',
      infoEmpty:   'Tidak ada data',
      infoFiltered:'(difilter dari _MAX_ total)',
      paginate:    { first: '«', last: '»', next: '›', previous:
'‹' },
      emptyTable:  '<div class="py-4 text-center text-muted"><i
class="fas fa-inbox fa-2x mb-2 d-block"></i>Belum ada data
mahasiswa</div>',
```

```
      zeroRecords: '<div class="py-3 text-center text-muted"><i
class="fas fa-search me-2"></i>Tidak ada data yang cocok</div>'
    },
    pageLength: 10,
    responsive: true,
    order:      [[1, 'asc']],
    dom:        '<"row mb-3"<"col-sm-6"l><"col-sm-6"f>>rtip'
  });

  // ── Tombol Refresh
──────────────────────────────────────────────────
  $('#btnRefresh').on('click', function () {
    const $btn = $(this);
    $btn.html('<i class="fas fa-arrows-rotate fa-spin me-
1"></i>Memuat...').prop('disabled', true);
    mahasiswaTable.ajax.reload(function () {
      $btn.html('<i class="fas fa-arrows-rotate me-1"></i>Muat
Ulang').prop('disabled', false);
      toastr.info('Data berhasil dimuat ulang');
    }, false);
  });

  // ── Tombol Edit
──────────────────────────────────────────────────
  $('#mahasiswaTable').on('click', '.btn-edit', function () {
    loadForEdit($(this).data('id'));
  });

  // ── Tombol Hapus
──────────────────────────────────────────────────
  $('#mahasiswaTable').on('click', '.btn-delete', function () {
    deleteTargetId = $(this).data('id');
    $('#deleteStudentName').text($(this).data('nama'));
    new
bootstrap.Modal(document.getElementById('deleteModal')).show();
  });

  // ── Konfirmasi Hapus
──────────────────────────────────────────────────
  $('#btnConfirmDelete').on('click', function () {
    if (!deleteTargetId) return;
    const $btn = $(this);
    $btn.html('<i class="fas fa-spinner fa-spin me-
1"></i>Menghapus...').prop('disabled', true);

    api.remove(deleteTargetId)
      .done(function (res) {
        bootstrap.Modal.getInstance(document.getElementById('delete
Modal')).hide();
        mahasiswaTable.ajax.reload(null, false);
        toastr.success(res.message || 'Data berhasil dihapus');
        deleteTargetId = null;
      })
      .fail(xhr => toastr.error(xhr.responseJSON?.error || 'Gagal
menghapus data'))
      .always(() => $btn.html('<i class="fas fa-trash-can me-
1"></i>Hapus').prop('disabled', false));
  });

  // ── Simpan Edit
──────────────────────────────────────────────────
```

```
  $('#btnSaveEdit').on('click', function () {
    const id = $('#editId').val();
    if (!id) return;

    let valid = true;
    ['editNim','editNama','editProdi','editAngkatan','editSemester'
,'editIpk','editEmail']
      .forEach(f => {
        if (!$('#' + f).val().trim()) {
          $('#' + f).addClass('is-invalid'); valid = false;
        } else {
          $('#' + f).removeClass('is-invalid');
        }
      });

    if (!valid) { showEditAlert('Harap isi semua field yang wajib
diisi.', 'danger'); return; }

    const ipkVal = parseFloat($('#editIpk').val());
    if (ipkVal < 0 || ipkVal > 4) { showEditAlert('IPK harus antara
0.00 dan 4.00', 'danger'); return; }

    const $btn = $(this);
    $btn.html('<i class="fas fa-spinner fa-spin me-
1"></i>Menyimpan...').prop('disabled', true);

    api.update(id, {
      nim:      $('#editNim').val().trim(),
      nama:     $('#editNama').val().trim(),
      prodi:    $('#editProdi').val(),
      angkatan: $('#editAngkatan').val(),
      semester: $('#editSemester').val(),
      ipk:      $('#editIpk').val(),
      email:    $('#editEmail').val().trim(),
      status:   $('#editStatus').val()
    })
      .done(function (res) {
        bootstrap.Modal.getInstance(document.getElementById('editMo
dal')).hide();
        mahasiswaTable.ajax.reload(null, false);
        toastr.success(res.message || 'Data berhasil diperbarui');
      })
      .fail(xhr => showEditAlert(xhr.responseJSON?.error || 'Gagal
memperbarui data', 'danger'))
      .always(() => $btn.html('<i class="fas fa-floppy-disk me-
2"></i>Simpan Perubahan').prop('disabled', false));
  });

  $('#editModal input, #editModal select').on('change input',
function () {
    $(this).removeClass('is-invalid');
  });
});

// ── Load data ke modal edit
──────────────────────────────────────
function loadForEdit(id) {
  api.getById(id)
    .done(function (d) {
      $('#editId').val(d.id);
      $('#editNim').val(d.nim);
```

```
      $('#editNama').val(d.nama);
      $('#editProdi').val(d.prodi);
      $('#editAngkatan').val(d.angkatan);
      $('#editSemester').val(d.semester);
      $('#editIpk').val(d.ipk);
      $('#editEmail').val(d.email);
      $('#editStatus').val(d.status);
      $('#editAlertArea').html('');
      new
bootstrap.Modal(document.getElementById('editModal')).show();
    })
    .fail(() => toastr.error('Gagal memuat data mahasiswa'));
}

function showEditAlert(msg, type) {
  $('#editAlertArea').html(
    `<div class="alert alert-${type} py-2 small rounded-3 mb-3">
       <i class="fas fa-circle-xmark me-2"></i>${msg}
     </div>`
  );
}
```

Source code form.js(Utils)

```
// form.js — Validasi form & AJAX submit (CREATE)

$(document).ready(function () {

  // ============================
  // jQuery Validate Plugin
  // ============================
  $('#studentForm').validate({
    rules: {
      nim:      { required: true, minlength: 5, maxlength: 10 },
      nama:     { required: true, minlength: 3 },
      email:    { required: true, email: true },
      prodi:    { required: true },
      angkatan: { required: true },
      semester: { required: true },
      ipk:      { required: true, min: 0, max: 4, number: true }
    },
    messages: {
      nim:      { required: 'NIM wajib diisi', minlength: 'Minimal
5 karakter', maxlength: 'Maksimal 10 karakter' },
      nama:     { required: 'Nama wajib diisi', minlength: 'Minimal
3 karakter' },
      email:    { required: 'Email wajib diisi', email: 'Format
email tidak valid' },
      prodi:    { required: 'Program studi wajib dipilih' },
      angkatan: { required: 'Angkatan wajib dipilih' },
      semester: { required: 'Semester wajib dipilih' },
      ipk:      { required: 'IPK wajib diisi', min: 'Minimal 0.00',
max: 'Maksimal 4.00', number: 'Harus berupa angka' }
    },
    errorElement: 'div',
    errorClass: 'invalid-feedback',
    highlight: function (el) { $(el).addClass('is-
invalid').removeClass('is-valid'); },
    unhighlight: function (el) { $(el).removeClass('is-
invalid').addClass('is-valid'); },
    errorPlacement: function (error, element) {
```

```
      if (element.parent().hasClass('input-group'))
error.insertAfter(element.parent());
      else error.insertAfter(element);
    },
    submitHandler: function () { submitForm(); }
  });

  // ============================
  // Submit via AJAX (POST — CREATE)
  // ============================
  function submitForm() {
    const $btn = $('#btnSubmit');
    $btn.html('<i class="fas fa-spinner fa-spin me-
2"></i>Menyimpan...').prop('disabled', true);
    $('#formAlertArea').html('');

    const payload = {
      nim:      $('#nim').val().trim(),
      nama:     $('#nama').val().trim(),
      email:    $('#email').val().trim(),
      prodi:    $('#prodi').val(),
      angkatan: $('#angkatan').val(),
      semester: $('#semester').val(),
      ipk:      $('#ipk').val(),
      status:   $('input[name="status"]:checked').val() || 'Aktif'
    };

    $.ajax({
      url: '/api/mahasiswa',
      type: 'POST',
      contentType: 'application/json',
      data: JSON.stringify(payload),
      success: function (res) {
        toastr.success(res.message);
        $('#formAlertArea').html(
          `<div class="alert alert-success d-flex align-items-
center gap-2 rounded-3">
             <i class="fas fa-circle-check"></i>
             <div><strong>Berhasil!</strong> Mahasiswa
<strong>${res.data.nama}</strong> telah terdaftar.
             <a href="/data" class="alert-link">Lihat rekap data
→</a></div>
           </div>`
        );
        resetForm();
        $('#previewSection').slideUp(200);
      },
      error: function (xhr) {
        const msg = xhr.responseJSON ? xhr.responseJSON.error :
'Terjadi kesalahan pada server';
        toastr.error(msg);
        $('#formAlertArea').html(
          `<div class="alert alert-danger d-flex align-items-center
gap-2 rounded-3">
             <i class="fas fa-circle-xmark"></i>
             <div><strong>Gagal!</strong> ${msg}</div>
           </div>`
        );
      },
      complete: function () {
```

```
        $btn.html('<i class="fas fa-floppy-disk me-
2"></i>Simpan').prop('disabled', false);
      }
    });
  }

  // ============================
  // Tombol Pratinjau (jQuery slideDown + animate)
  // ============================
  $('#btnPreview').on('click', function () {
    if (!$('#studentForm').valid()) {
      toastr.warning('Lengkapi semua field yang wajib diisi
terlebih dahulu');
      return;
    }
    const nim      = $('#nim').val();
    const nama     = $('#nama').val();
    const email    = $('#email').val();
    const prodi    = $('#prodi').val();
    const angkatan = $('#angkatan').val();
    const semester = $('#semester').val();
    const ipk      = parseFloat($('#ipk').val()).toFixed(2);
    const status   = $('input[name="status"]:checked').val() ||
'Aktif';

    const html = `
      <div class="col-6 col-md-3">
        <div class="preview-item-label">NIM</div>
        <div class="preview-item-val">${nim}</div>
      </div>
      <div class="col-6 col-md-5">
        <div class="preview-item-label">Nama</div>
        <div class="preview-item-val">${nama}</div>
      </div>
      <div class="col-12 col-md-4">
        <div class="preview-item-label">Email</div>
        <div class="preview-item-val" style="font-
size:0.8rem">${email}</div>
      </div>
      <div class="col-6 col-md-4">
        <div class="preview-item-label">Program Studi</div>
        <div class="preview-item-val">${prodi}</div>
      </div>
      <div class="col-6 col-md-2">
        <div class="preview-item-label">Angkatan</div>
        <div class="preview-item-val">${angkatan}</div>
      </div>
      <div class="col-6 col-md-2">
        <div class="preview-item-label">Semester</div>
        <div class="preview-item-val">Sem-${semester}</div>
      </div>
      <div class="col-6 col-md-2">
        <div class="preview-item-label">IPK</div>
        <div class="preview-item-val" style="color:var(--
warning)">${ipk}</div>
      </div>
      <div class="col-6 col-md-2">
        <div class="preview-item-label">Status</div>
        <div class="preview-item-val">${status}</div>
      </div>`;
    $('#previewContent').html(html);
```

```
    $('#previewSection').slideDown(300);
    $('html, body').animate({ scrollTop:
$('#previewSection').offset().top - 100 }, 500);
  });

  // ============================
  // Tombol Reset
  // ============================
  $('#btnReset').on('click', function () {
    resetForm();
    $('#previewSection').slideUp(200);
    $('#formAlertArea').html('');
    toastr.info('Form berhasil direset');
  });

  function resetForm() {
    $('#studentForm')[0].reset();
    $('#studentForm .is-invalid, #studentForm .is-
valid').removeClass('is-invalid is-valid');
    $('#statusAktif').prop('checked', true);
  }

  // ============================
  // Real-time input filtering (jQuery)
  // ============================
  $('#nim').on('input', function () {
    $(this).val($(this).val().replace(/\D/g, ''));
  });

  $('#ipk').on('input', function () {
    const v = parseFloat($(this).val());
    if (v > 4) $(this).val(4);
    if (v < 0) $(this).val(0);
  });

  // Highlight input-group-text icon on focus
  $('input, select').on('focus', function () {
    $(this).closest('.input-group').find('.input-group-
text').addClass('ig-focused');
  }).on('blur', function () {
    $(this).closest('.input-group').find('.input-group-
text').removeClass('ig-focused');
  });
});
```

Source code main.js (utils)

```
// main.js — Konfigurasi global SIAKAD

// Toastr options
if (typeof toastr !== 'undefined') {
  toastr.options = {
    closeButton: true,
    progressBar: true,
    positionClass: 'toast-top-right',
    timeOut: 3500,
    extendedTimeOut: 1500,
    showMethod: 'slideDown',
    hideMethod: 'slideUp'
  };
```

```
}

// Auto-set active navbar link based on current path
$(document).ready(function () {
  const path = window.location.pathname;
  $('.navbar-nav .nav-link').each(function () {
    const href = $(this).attr('href');
    if (href === path) {
      $(this).addClass('active');
    } else if (href !== '/') {
      $(this).removeClass('active');
    } else if (path === '/') {
      $(this).addClass('active');
    } else {
      $(this).removeClass('active');
    }
  });
});
```

Source code table.js(utils)

```
// table.js — DataTable CRUD (Read, Update, Delete)
let mahasiswaTable;
let deleteTargetId = null;

$(document).ready(function () {

  // ============================
  // Inisialisasi jQuery DataTable (JSON AJAX)
  // ============================
  mahasiswaTable = $('#mahasiswaTable').DataTable({
    ajax: {
      url: '/api/mahasiswa',
      type: 'GET',
      dataSrc: 'data'
    },
    columns: [
      {
        data: null,
        render: function (data, type, row, meta) {
          return '<span class="nim-mono">' + (meta.row + 1) +
'</span>';
        },
        orderable: false
      },
      {
        data: 'nim',
        render: function (d) { return '<span class="nim-mono">' + d
+ '</span>'; }
      },
      {
        data: 'nama',
        render: function (d) { return '<strong>' + d + '</strong>';
}
      },
      { data: 'prodi' },
      {
        data: 'angkatan',
        render: function (d) { return '<span class="nim-mono">' + d
+ '</span>'; }
      },
```

```
      {
        data: 'semester',
        render: function (d) { return '<span class="nim-mono">Sem-'
+ d + '</span>'; }
      },
      {
        data: 'ipk',
        render: function (d) {
          const v = parseFloat(d);
          const cls = v >= 3.5 ? 'ipk-high' : v >= 3.0 ? 'ipk-mid'
: 'ipk-low';
          return '<span class="ipk-display ' + cls + '">' +
v.toFixed(2) + '</span>';
        }
      },
      {
        data: 'status',
        render: function (d) {
          const map = { 'Aktif':'badge-aktif','Cuti':'badge-
cuti','Lulus':'badge-lulus','Keluar':'badge-keluar' };
          return '<span class="status-badge ' + (map[d]||'badge-
aktif') + '">' + d + '</span>';
        }
      },
      {
        data: null,
        orderable: false,
        render: function (data, type, row) {
          return `<div class="d-flex gap-1">
            <button class="btn-action btn-edit" data-id="${row.id}"
title="Edit data">
              <i class="fas fa-pen"></i>
            </button>
            <button class="btn-action btn-delete" data-
id="${row.id}" data-nama="${row.nama}" title="Hapus data">
              <i class="fas fa-trash-can"></i>
            </button>
          </div>`;
        }
      }
    ],
    language: {
      search: 'Cari:',
      lengthMenu: 'Tampilkan _MENU_ data',
      info: 'Menampilkan _START_–_END_ dari _TOTAL_ mahasiswa',
      infoEmpty: 'Tidak ada data',
      infoFiltered: '(difilter dari _MAX_ total)',
      paginate: { first: '«', last: '»', next: '›', previous: '‹'
},
      emptyTable: '<div class="py-4 text-center text-muted"><i
class="fas fa-inbox fa-2x mb-2 d-block"></i>Belum ada data
mahasiswa terdaftar</div>',
      zeroRecords: '<div class="py-3 text-center text-muted"><i
class="fas fa-search me-2"></i>Tidak ada data yang cocok</div>'
    },
    pageLength: 10,
    responsive: true,
    order: [[1, 'asc']],
    dom: '<"row mb-3"<"col-sm-6"l><"col-sm-6"f>>rtip'
  });
```

```
  // ============================
  // Tombol Muat Ulang (Refresh)
  // ============================
  $('#btnRefresh').on('click', function () {
    const $btn = $(this);
    $btn.html('<i class="fas fa-arrows-rotate fa-spin me-
1"></i>Memuat...').prop('disabled', true);
    mahasiswaTable.ajax.reload(function () {
      $btn.html('<i class="fas fa-arrows-rotate me-1"></i>Muat
Ulang').prop('disabled', false);
      toastr.info('Data berhasil dimuat ulang');
    }, false);
  });

  // ============================
  // Klik Tombol Edit (event delegation)
  // ============================
  $('#mahasiswaTable').on('click', '.btn-edit', function () {
    const id = $(this).data('id');
    loadForEdit(id);
  });

  // ============================
  // Klik Tombol Hapus
  // ============================
  $('#mahasiswaTable').on('click', '.btn-delete', function () {
    deleteTargetId = $(this).data('id');
    const nama = $(this).data('nama');
    $('#deleteStudentName').text(nama);
    new
bootstrap.Modal(document.getElementById('deleteModal')).show();
  });

  // ============================
  // Konfirmasi Hapus (DELETE)
  // ============================
  $('#btnConfirmDelete').on('click', function () {
    if (!deleteTargetId) return;
    const $btn = $(this);
    $btn.html('<i class="fas fa-spinner fa-spin me-
1"></i>Menghapus...').prop('disabled', true);

    $.ajax({
      url: '/api/mahasiswa/' + deleteTargetId,
      type: 'DELETE',
      success: function (res) {
        bootstrap.Modal.getInstance(document.getElementById('delete
Modal')).hide();
        mahasiswaTable.ajax.reload(null, false);
        toastr.success(res.message || 'Data berhasil dihapus');
        deleteTargetId = null;
      },
      error: function (xhr) {
        toastr.error(xhr.responseJSON ? xhr.responseJSON.error :
'Gagal menghapus data');
      },
      complete: function () {
        $btn.html('<i class="fas fa-trash-can me-
1"></i>Hapus').prop('disabled', false);
      }
    });
```

```
  });

  // ============================
  // Simpan Perubahan Edit (PUT — UPDATE)
  // ============================
  $('#btnSaveEdit').on('click', function () {
    const id = $('#editId').val();
    if (!id) return;

    let valid = true;
    ['editNim','editNama','editProdi','editAngkatan','editSemester'
,'editIpk','editEmail'].forEach(function (f) {
      if (!$('#' + f).val().trim()) {
        $('#' + f).addClass('is-invalid'); valid = false;
      } else {
        $('#' + f).removeClass('is-invalid');
      }
    });
    if (!valid) { showEditAlert('Harap isi semua field yang wajib
diisi.', 'danger'); return; }

    const ipkVal = parseFloat($('#editIpk').val());
    if (ipkVal < 0 || ipkVal > 4) { showEditAlert('IPK harus antara
0.00 dan 4.00', 'danger'); return; }

    const $btn = $(this);
    $btn.html('<i class="fas fa-spinner fa-spin me-
1"></i>Menyimpan...').prop('disabled', true);

    $.ajax({
      url: '/api/mahasiswa/' + id,
      type: 'PUT',
      contentType: 'application/json',
      data: JSON.stringify({
        nim:      $('#editNim').val().trim(),
        nama:     $('#editNama').val().trim(),
        prodi:    $('#editProdi').val(),
        angkatan: $('#editAngkatan').val(),
        semester: $('#editSemester').val(),
        ipk:      $('#editIpk').val(),
        email:    $('#editEmail').val().trim(),
        status:   $('#editStatus').val()
      }),
      success: function (res) {
        bootstrap.Modal.getInstance(document.getElementById('editMo
dal')).hide();
        mahasiswaTable.ajax.reload(null, false);
        toastr.success(res.message || 'Data berhasil diperbarui');
      },
      error: function (xhr) {
        showEditAlert(xhr.responseJSON ? xhr.responseJSON.error :
'Gagal memperbarui data', 'danger');
      },
      complete: function () {
        $btn.html('<i class="fas fa-floppy-disk me-2"></i>Simpan
Perubahan').prop('disabled', false);
      }
    });
  });

  // Reset invalid class on change
```

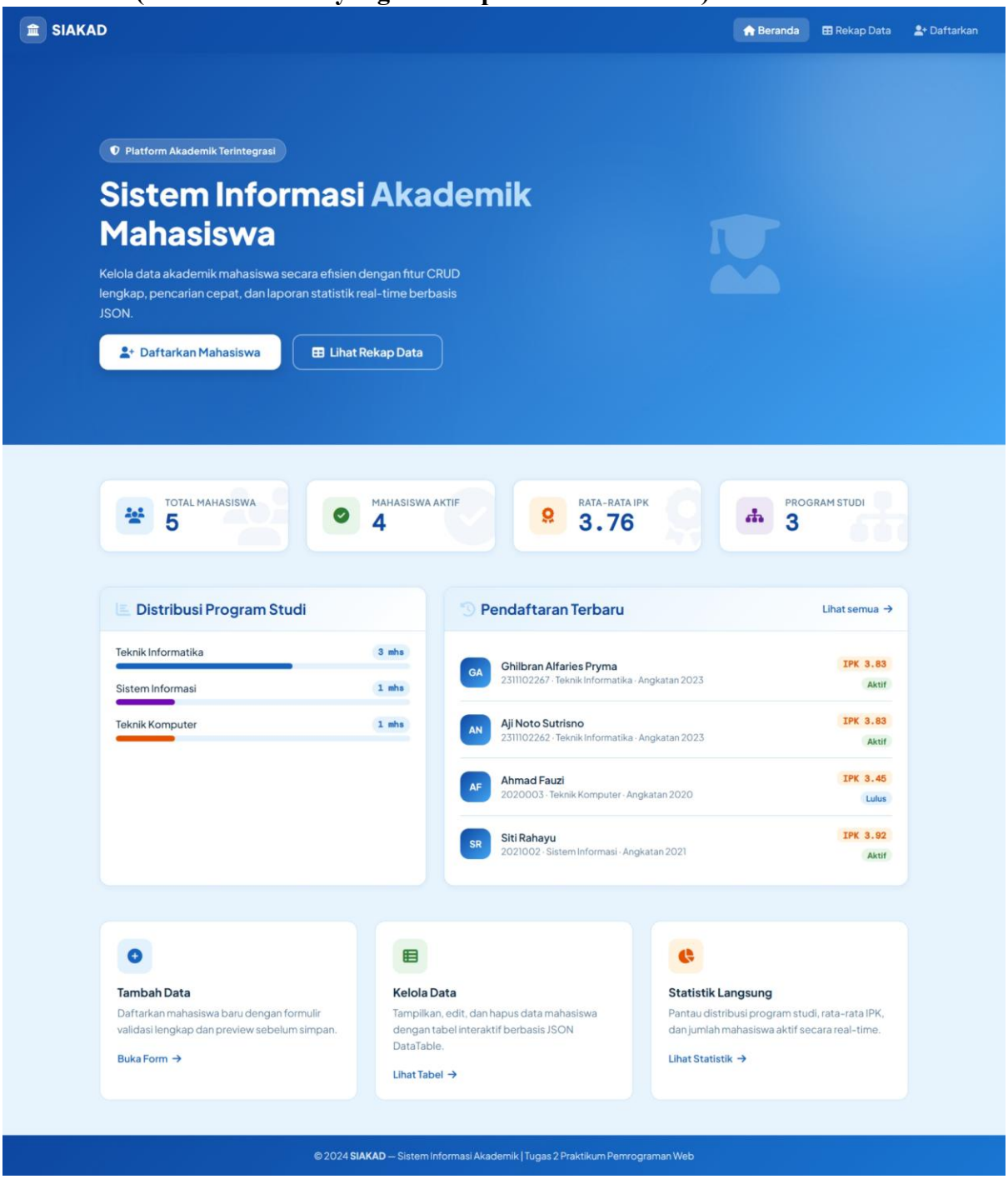
```
  $('#editModal input, #editModal select').on('change input',
function () {
    $(this).removeClass('is-invalid');
  });
});

// Load data ke modal edit (READ by ID)
function loadForEdit(id) {
  $.getJSON('/api/mahasiswa/' + id, function (d) {
    $('#editId').val(d.id);
    $('#editNim').val(d.nim);
    $('#editNama').val(d.nama);
    $('#editProdi').val(d.prodi);
    $('#editAngkatan').val(d.angkatan);
    $('#editSemester').val(d.semester);
    $('#editIpk').val(d.ipk);
    $('#editEmail').val(d.email);
    $('#editStatus').val(d.status);
    $('#editAlertArea').html('');
    new
bootstrap.Modal(document.getElementById('editModal')).show();
  }).fail(function () {
    toastr.error('Gagal memuat data mahasiswa');
  });
}

function showEditAlert(msg, type) {
  $('#editAlertArea').html(
    `<div class="alert alert-${type} py-2 small rounded-3 mb-3"><i
class="fas fa-circle-xmark me-2"></i>${msg}</div>`
  );
}
```

Source code mahasiswaRoute.js (routes)

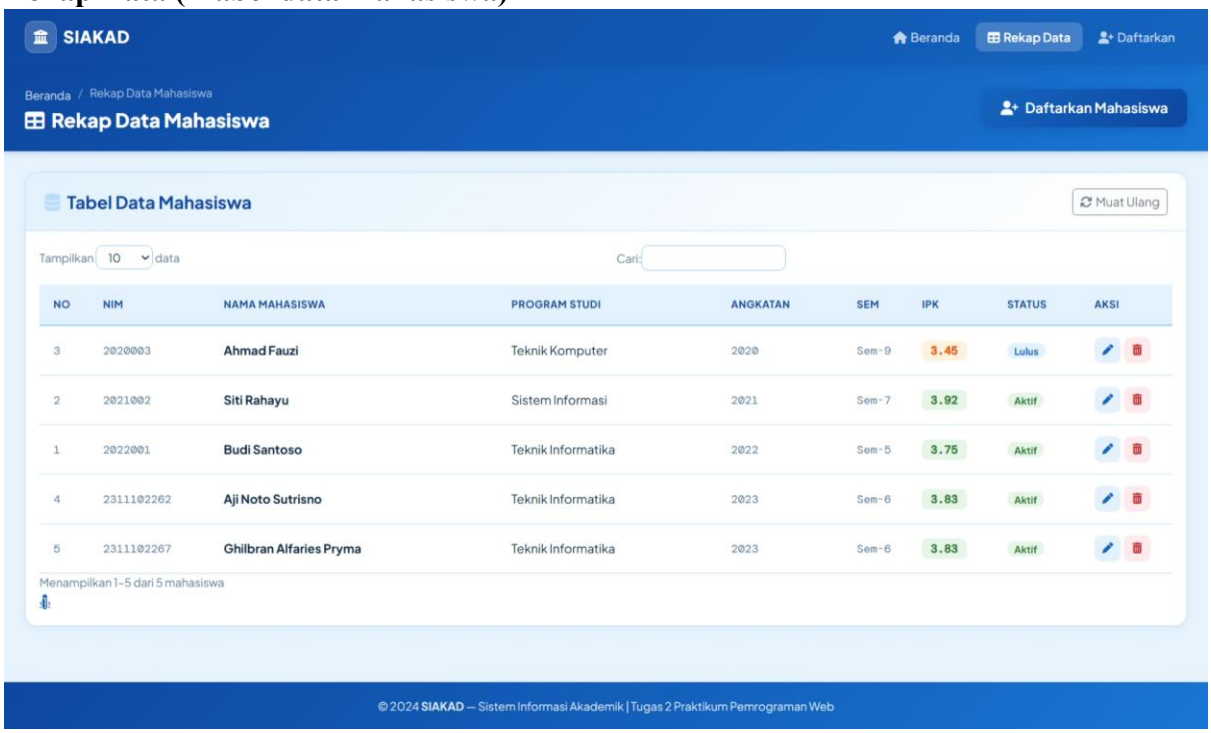
```
const express    = require('express');
const router     = express.Router();
const controller = require('../controllers/mahasiswaController');

router.get('/',        controller.getAll);
router.get('/stats',   controller.getStats);
router.get('/:id',     controller.getById);
router.post('/',       controller.create);
router.put('/:id',     controller.update);
router.delete('/:id',  controller.remove);

module.exports = router;
```

7. **Output Program**

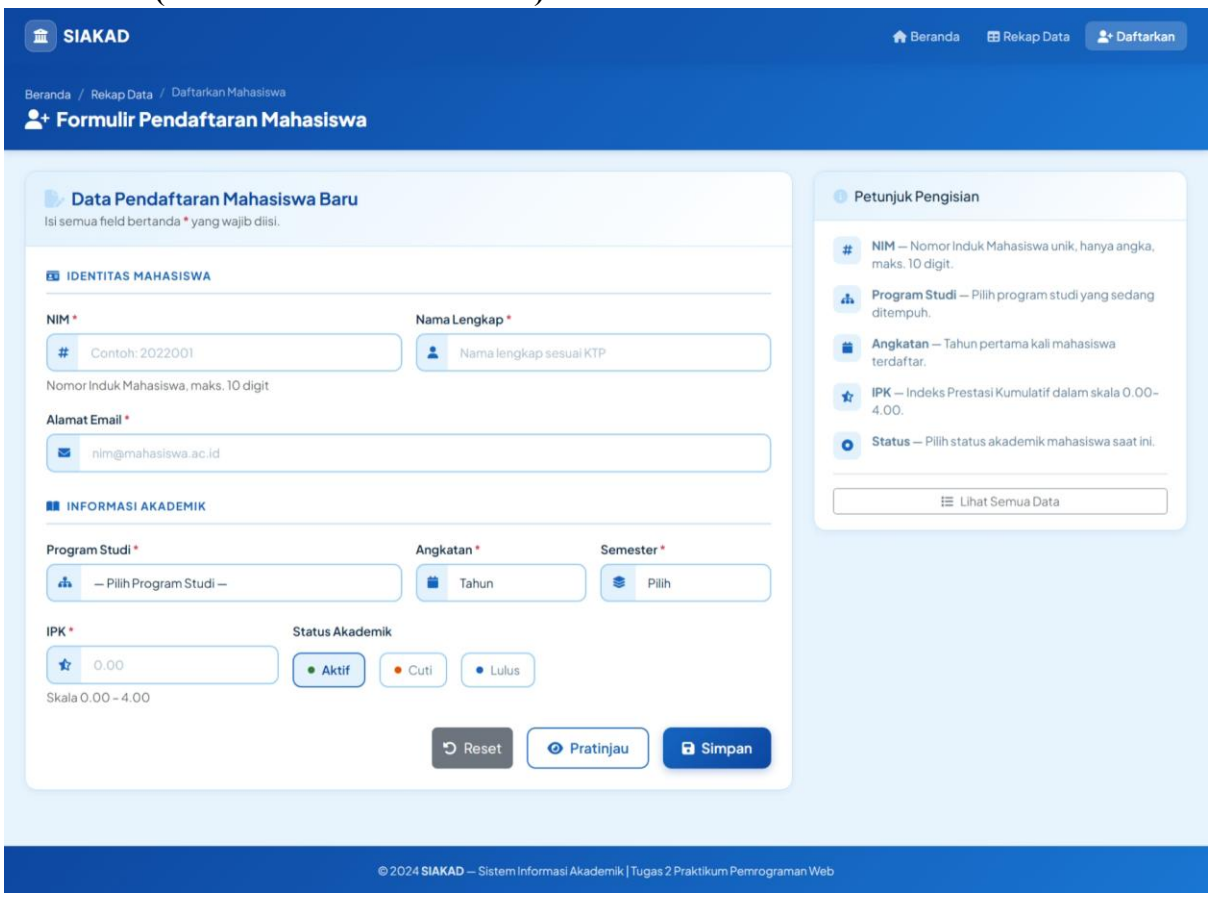
Dashboard (Halaman utama yang menampilkan keseluruhan)



Rekap Data (Tabel data mahasiswa)



Daftarkan (Menambahkan Mahasiswa)



Akun Github : Webprogram/Ramadhan at main · Jianst-21/Webprogram